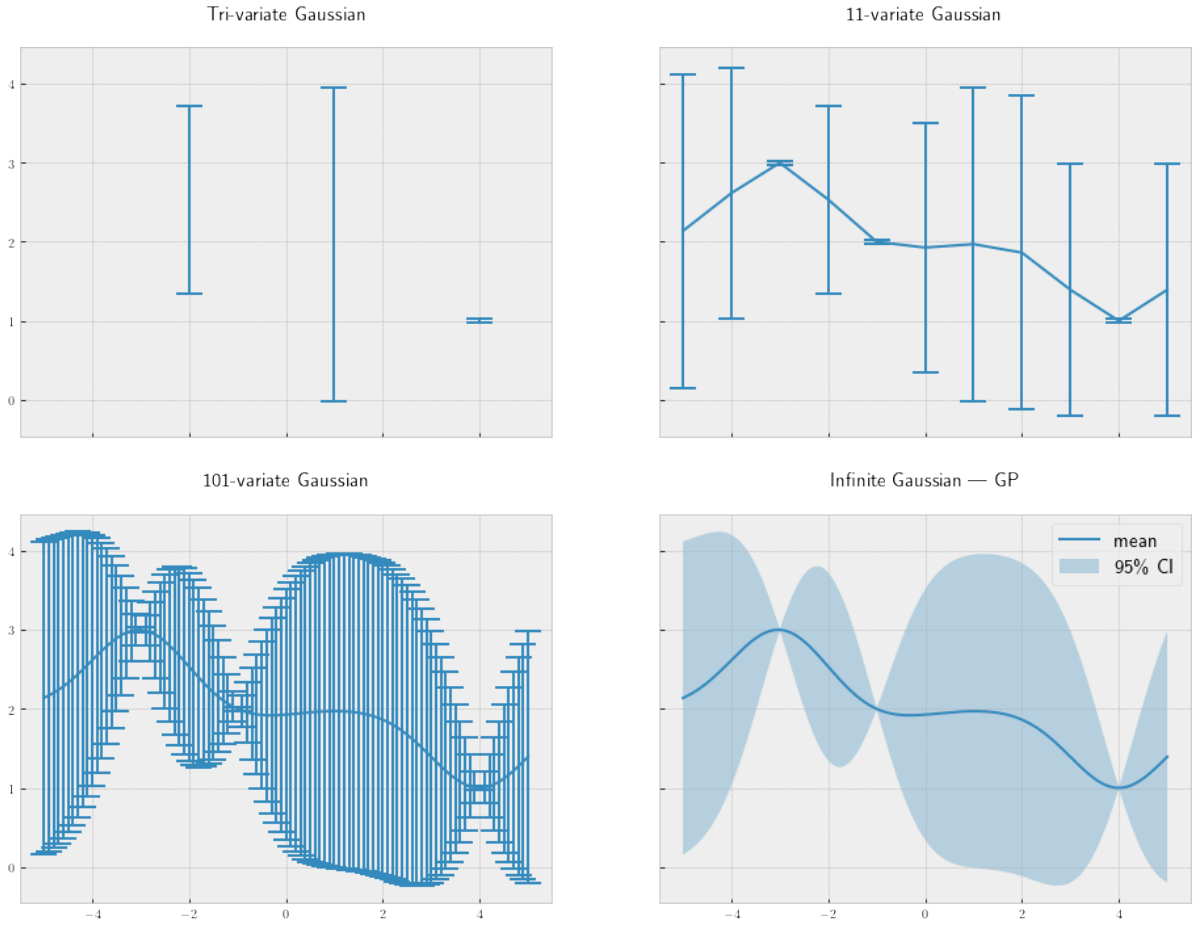


Figure 2.13 Parallel coordinates plots of different Gaussian distributions. Any finite subset of a GP is a multivarate Gaussian. As the number of variables approaches infinity, we obtain a GP and can make predictions everywhere in the domain.

This is illustrated in Figure 2.13. The first three panels show in parallel coordinates plots a tri-variate Gaussian for $[f(-2)\ f(0.2)\ f(12)\ f(4)]$, a 11-variate Gaussian for $[f(-4.5)\ f(-4)\ \ldots\ f(4)\ f(4.5)]^{\mathrm{T}}$, and a 101-variate Gaussian across a denser grid. Finally, in the last panel, we have infinitely many variables, which gives us a GP.

Since we are now in infinite dimensions, it doesn't make sense for us to talk about the mean vector and the covariance matrix anymore. Instead, what we have with a GP is a mean *function* and a covariance *function*, but the respective roles of these two objects are still the same as with a multivariate Gaussian.

- First, the mean function, which takes in one input x, computes the expectation of the function value $f(x)$.
- Second, the covariance function takes into two inputs x and x' and computes the covariance between the two variables $f(x)$ and $f(x')$. If x' is the same as x, then this covariance value is simply the variance of the normal distribution of $f(x)$. If x' is different from x, the covariance denotes the correlation between the two variables.

As the mean and the covariance are functions, we are no longer tied to a fixed number of

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